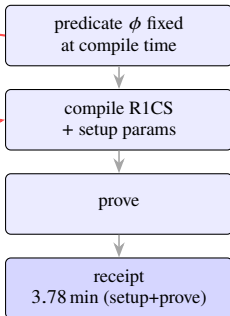


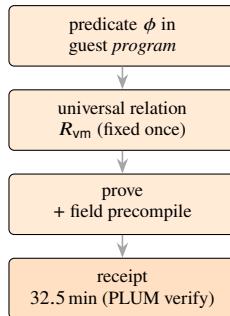
Static circuit (Aurora)

ϕ change
 $\Rightarrow$ recompile
 $N_{\text{circ}}, N_{\text{param}} \geq 1$



General-purpose zkVM

ϕ change
 $\Rightarrow$ guest edit
 $N_{\text{circ}} = N_{\text{param}} = 0$



Crossover: static cheaper while update rate $U/P < (t_{\text{zkVM}}^P - t_{\text{static}}^P)/R_{\text{static}}$; above it, the zkVM's $N_{\text{circ}} = 0$ wins over a churning deployment.